

Polynomial Regression: Predictive Modeling for Geothermal Expansion

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<https://github.com/dhyanspatel49/Assignment-Polynomial-Regression>

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Abstract

This report details the development of polynomial regression models to optimize a multi-stage geothermal power plant (Phase 1) and map a subterranean thermal reservoir (Phase 2). By employing polynomial feature expansion, feature scaling, L2 regularization (Ridge Regression), and K-Fold Cross-Validation, optimal polynomial degrees were identified for both unique datasets. The resulting models successfully capture complex non-linear relationships while rigorously guarding against multicollinearity and overfitting.

1 Introduction

The objective of this assignment is to predict a continuous target variable y from a given set of input variables across two distinct problem statements. The constraints require the exclusive use of polynomial regression. Due to the high dimensionality generated by polynomial feature expansion, a regularized approach was adopted to ensure mathematical stability and high predictive accuracy on unseen test data.

2 Methodology and Techniques

To construct robust predictive models, a unified data pipeline was developed and applied to both problem domains. The pipeline incorporates the following sequential steps:

2.1 Polynomial Feature Expansion

For a given set of features, polynomial expansion generates all polynomial combinations up to a specified maximum degree. While necessary for capturing complex non-linear geometric and thermodynamic relationships, this process significantly increases dimensionality (e.g., a degree-10 expansion on 6 variables yields 8,007 features).

2.2 Feature Scaling (Standardization)

Because polynomial terms compound exponentially, high-degree features can easily trigger floating-point overflow and dominate the regression algorithm. To prevent this, a

`StandardScaler` was applied immediately following feature expansion to center the mean at 0 and scale the variance to 1 across all polynomial features.

2.3 Regularized Polynomial Regression (Ridge)

Standard Ordinary Least Squares (OLS) regression becomes highly unstable when dealing with thousands of correlated polynomial features, often resulting in singular matrices where the determinant of $X^T X$ approaches zero. To solve this, L2 Regularization (Ridge Regression) was integrated. The Ridge objective function minimizes both the Residual Sum of Squares (RSS) and the squared magnitude of coefficients:

$$Loss = \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \alpha \sum_{j=1}^p \beta_j^2 \quad (1)$$

By tuning the penalty parameter α via internal cross-validation (`RidgeCV`), the model retains the continuous physical interactions of the variables while artificially constraining the weights, thereby preventing the severe overfitting characteristic of high-degree OLS models.

2.4 K-Fold Cross-Validation for Degree Selection

To select the optimal polynomial degree without leaking test data, 5-Fold Cross-Validation (`KFold(n_splits=5)`) was utilized on the training sets. The algorithm evaluated degrees sequentially, recording the Mean Squared Error (MSE) and Coefficient of Determination (R^2). The degree yielding the highest R^2 across the validation folds was selected for final inference.

3 Phase 1: Power Plant Steam Turbine Optimization (var1)

3.1 Context and Approach

Phase 1 required predicting the Net Power Score (y) based on six turbine operational parameters (x_1 through x_6). Given the 6-dimensional input space, the polynomial degree was capped at 10 to ensure computational feasibility.

3.2 Results and Optimal Degree Selection

The cross-validation process revealed a clear bias-variance tradeoff, summarized in Table 1.

Table 1: Cross-Validation Metrics for Phase 1 (var1)

Degree	Mean R^2 Score	Mean MSE
1	0.0773	9.6362
2	0.7122	2.9597
3	0.9009	1.0147
4	0.9307	0.7094
5	0.9512	0.4965
6	0.9426	0.5886
7	0.9339	0.6759
8	0.9173	0.8510
9	0.9102	0.9236
10	0.8938	1.0986

Conclusion for Phase 1: The model performance improved consistently up to **Degree 5**, where it achieved a peak Mean R^2 of 0.9512 and minimized the MSE at 0.4965. Beyond Degree 5, the model began memorizing training noise (overfitting), leading to a decline in R^2 and an increase in MSE. Therefore, a 5th-degree polynomial was selected. Final predictions were generated and saved to `BT2024075_pred_var1.csv`.

4 Phase 2: Subterranean Thermal Reservoir Mapping (var2)

4.1 Context and Approach

Phase 2 involved mapping a Thermal Anomaly Score (y) using three 3D spatial coordinate offsets (x_1, x_2, x_3). Because the feature space only contained 3 variables, the maximum polynomial degree was safely extended to 20 to capture the highly complex subterranean geological structures.

4.2 Results and Optimal Degree Selection

The evaluation metrics across 20 degrees demonstrated a rapid initial improvement followed by a plateau maintained by the L2 regularization.

Table 2: Cross-Validation Metrics for Phase 2 (var2)

Degree	Mean R^2 Score	Mean MSE
1	0.1982	32.9530
2	0.5092	20.4104
3	0.7226	11.2399
...	...	...
7	0.9914	0.3508
8	0.9932	0.2797
9	0.9935	0.2675
10	0.9937	0.2601
11	0.9936	0.2641
15	0.9931	0.2871
20	0.9921	0.3287

Conclusion for Phase 2: The R^2 score climbed aggressively in the lower degrees, indicating a highly non-linear thermal gradient. The metrics reached absolute optimization at **Degree 10** (R^2 : 0.9937, MSE: 0.2601). While degrees 11 through 20 remained highly stable due to the Ridge penalty capping outlier coefficients, Degree 10 was selected as the mathematically optimal peak. Final predictions were strictly formatted and saved to BT2024075_pred_var2.csv.

5 Final Deliverables

The adopted methodology successfully navigated the geometric complexities and computational hazards of high-degree polynomial regression. The resulting predictive models and their generated datasets provide a robust, highly accurate foundation for optimizing surface turbine operations and guiding subterranean drilling strategies.